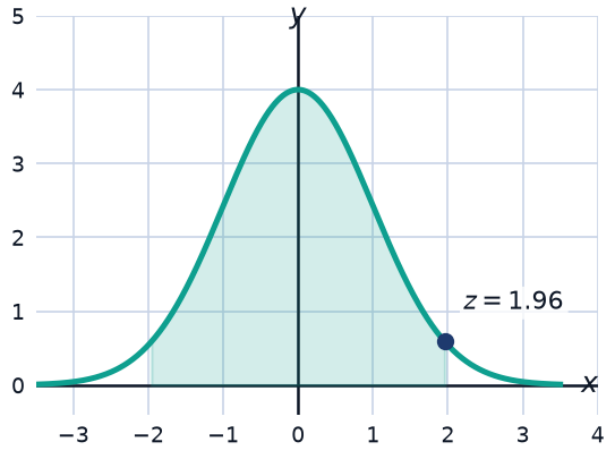
Systems of Linear Equations (AA HL)

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.05

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two independent random variables are normally distributed as $X \sim N(50, 16)$ and $Y \sim N(45, 9)$. Find the probability that $X > Y$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the points $A(1, -2, 3)$, $B(4, 0, 5)$, and $C(p, 1, 2)$. If the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are perpendicular, find the value of p.

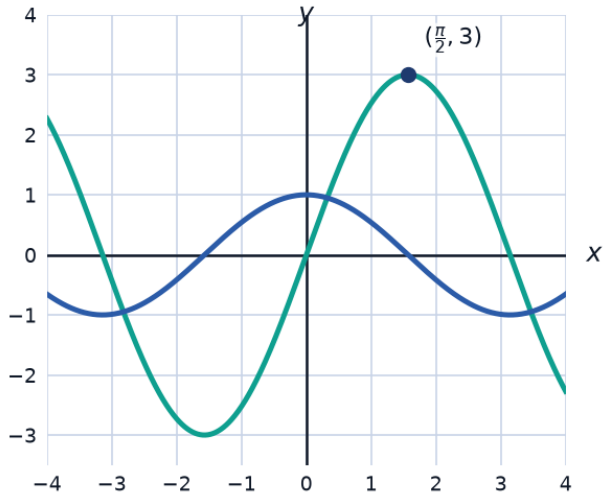
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A fair six-sided die is rolled. What is the probability of rolling a number greater than 4?

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Phone calls arrive at an emergency center according to a Poisson process with an average rate of 3 calls per minute. Find: (a) the probability that at least 2 calls arrive in a 1-minute period; (b) the probability that exactly 5 calls arrive in a 2-minute period.



5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A discrete random variable X takes values $0, 3, 4$ with probabilities $\frac{1}{10}, \frac{3}{10}, \frac{3}{5}$ respectively. Find $E(X)$.

6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The region bounded by $y = \ln x$, the line $y = 0$, the line $y = 1$, and the y-axis is rotated 2π radians about the y-axis. Find the exact volume of the resulting solid.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

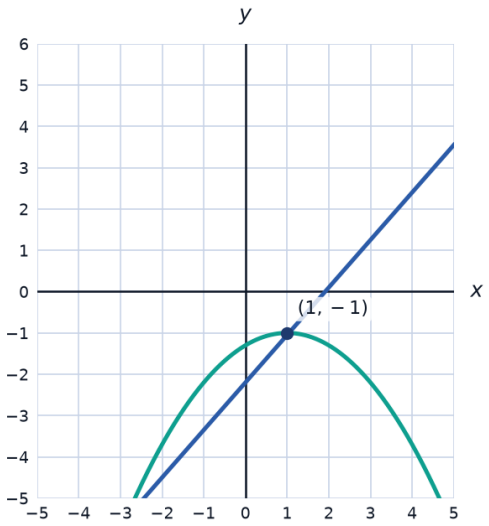
Consider the function $f(x) = \frac{1}{2}x^2 - 3x + 5$. The graph of $y = f(x)$ is a parabola. Find the x-coordinate of the vertex of the parabola.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find $\int xe^{2x}\,dx$ (in terms of x, with constant C).



9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the number of solutions of $\sin(2\theta) = \frac{1}{2}$ in the interval $0^\circ \le \theta < 360^\circ$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

How many solutions does $\sin\theta = \sin 30^\circ$ have in the interval $0^\circ \le \theta < 360^\circ$? Show your complete algebraic work.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A network has weighted adjacency matrix W. Explain how the minimum spanning tree algorithm (Kruskal's) differs from Dijkstra's algorithm.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A student receives the following scaled marks on IB AA HL assessment: Paper 1: 85/110, Paper 2: 92/110, Paper 3: 44/55, and IA: 16/20. Calculate the student's overall weighted percentage.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A continuous random variable X has probability density function $f(x) = 2e^{-2x}$ for $x \ge 0$ and 0 elsewhere. Find the cumulative distribution function $F(x)$ and determine the exact median of X.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the perpendicular distance from point $P(3, 1, -2)$ to plane $\pi : 2x - 2y + z = 5$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Find the volume of the solid generated when the region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ is revolved 2π radians about the x-axis.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Find the Maclaurin series for $\ln(1 + 2x)$ up to the x^3 term, and state its interval of convergence.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.04 Partial Fractions

[AA HL — Additional Topics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

[AI HL — Additional Topics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

[Overview · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

[Functions (HL Extensions) · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In row-reducing an augmented matrix, which row operation is NOT permitted?

a. Squaring all elements in a row

b. Swapping two rows

c. Multiplying a row by a non-zero scalar

d. Adding a multiple of one row to another row

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An augmented matrix in row echelon form has final row $[0, 0, 0 \mid 5]$. What does this indicate about the system?

a. The system is inconsistent (has no solutions)

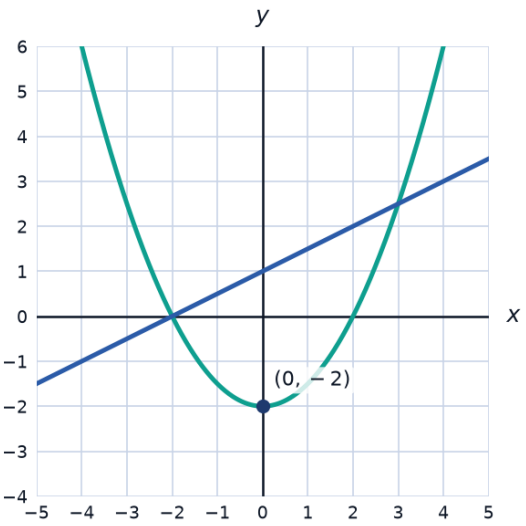
b. The system has infinitely many solutions

c. The system has a unique solution with $z = 5$

d. The system has trivial solutions only

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An augmented matrix in row echelon form has final row $[0, 0, 0 \mid 0]$ with 2 leading pivots across 3 variables. What does this signify?



a. The system has infinitely many solutions parameterized by 1 free variable

b. The system has a unique solution

c. The system has no solutions

d. The determinant of the matrix is 1

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the geometric configuration of three linear equations in three variables that possess a unique solution?

a. Three planes intersecting at a single unique point

b. Three planes intersecting along a common line

c. Three parallel non-intersecting planes

d. Three planes forming a triangular prism

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If three planes intersect along a single straight line, what type of system is this?

a. Consistent dependent system with infinitely many solutions

b. Inconsistent system with no solutions

c. Consistent independent system with 1 solution

d. Homogeneous system with trivial solution only

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Three planes form the faces of an open triangular prism with no common line or point. How is this system classified?

a. Inconsistent (no solutions)

b. Infinitely many solutions

c. Unique solution

d. Singular dependent

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Evaluate the 2×2 determinant $\det[[4, -3], [2, 5]]$.

a. 26

b. 14

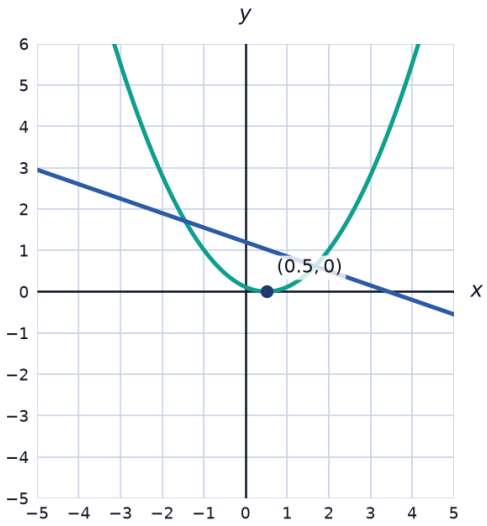
c. -26

d. 20

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Under what condition does a 3×3 system $A\mathbf{x} = \mathbf{b}$ have a unique solution?



a. $\det(A) \neq 0$

b. $\det(A) = 0$

c. $\mathbf{b} = \mathbf{0}$

d. $\text{Tr}(A) > 0$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $\det(A) = 0$ for a system $A\mathbf{x} = \mathbf{b}$, what can be concluded?

a. The system either has no solutions or infinitely many solutions

b. The system must have no solutions

c. The system must have infinitely many solutions

d. The system has exactly 2 solutions

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve the system: $x + y + z = 6$ and $2x - y + z = 3$. Parameterize the solution line using $z = \lambda$.

a. $x = 3 - \frac{2}{3}\lambda, y = 3 - \frac{1}{3}\lambda, z = \lambda$

b. $x = 3 + \lambda, y = 3 - \lambda, z = \lambda$

c. $x = 9 - 2\lambda, y = \lambda, z = \lambda$

d. $x = 1, y = 2, z = 3$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of k does the system $x + 2y = 4$ and $3x + 6y = k$ have infinitely many solutions?

a. $k = 12$

b. $k = 4$

c. $k = 0$

d. $k = 6$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If the homogeneous system $A\mathbf{x} = \mathbf{0}$ has non-trivial solutions (solutions other than $\mathbf{x} = \mathbf{0}$), what must be true about $\det(A)$?

a. $\det(A) = 0$

b. $\det(A) > 0$

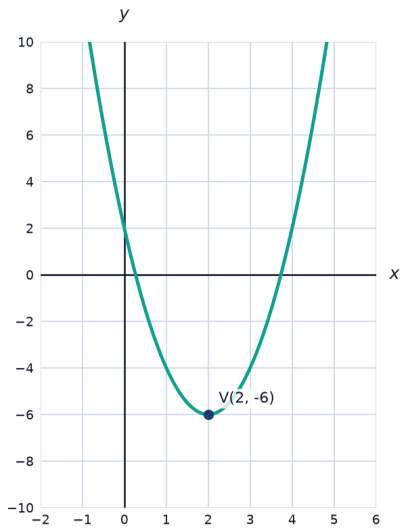
c. $\det(A) = 1$

d. $\det(A) < 0$

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagram shows the graph of the function $f(x) = 2x^2 - 8x + k$. The graph intersects the x-axis at points A and B, and the vertex of the parabola is at point V. Given that the x-coordinate of the vertex is 2, and the y-coordinate of the vertex is -6, find the value of k.



a. 1

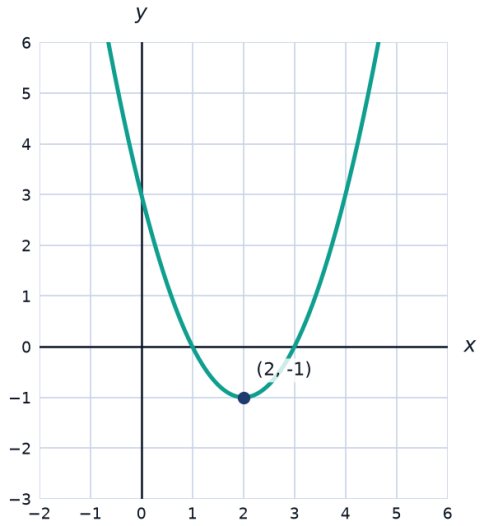
b. -2

c. 2

d. -1

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is a parabola. Find the coordinates of the vertex of the parabola.



a. (2, -1)

b. (-2, 1)

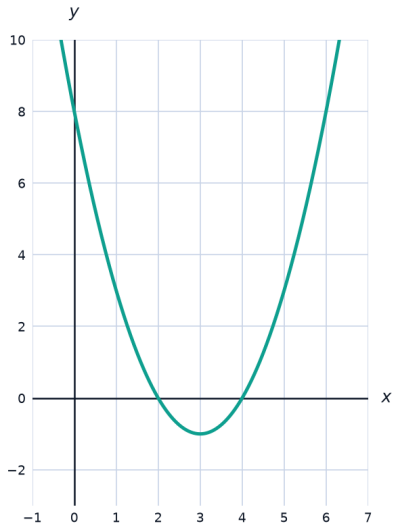
c. (2, 1)

d. (-2, -1)

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $f(x) = x^2 - 6x + 8$. The graph of $y = f(x)$ intersects the x-axis at points A and B, and the y-axis at point C. Find the length of the line segment AB.



a. 2 units

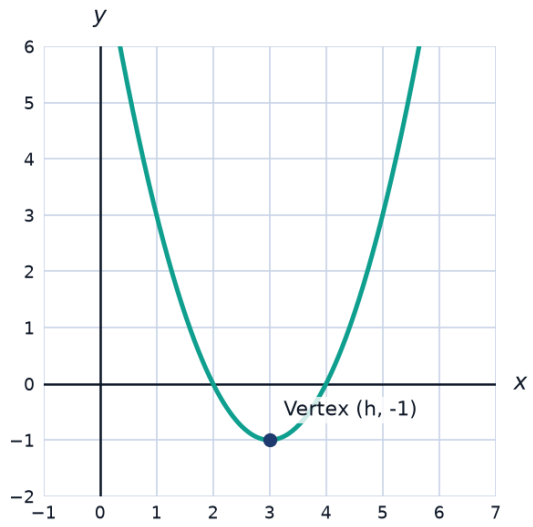
b. 3 units

c. 4 units

d. 6 units

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The diagram shows the graph of the function $f(x) = x^2 - 6x + k$. The vertex of the parabola is at the point $(h, -1)$. Determine the value of k.



a. 8

b. 7

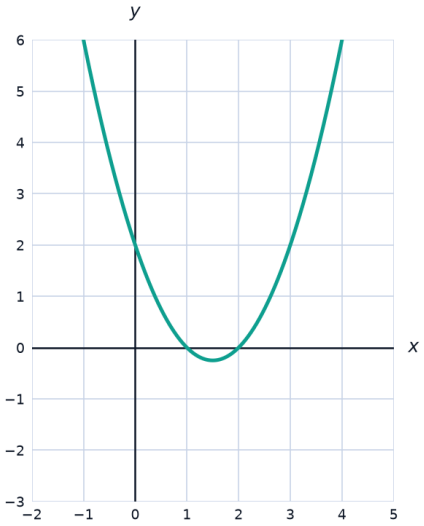
c. 6

d. 5

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37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Consider the function $f(x) = x^2 - 3x + 2$. The graph of $y = f(x)$ is a parabola. Find the x-coordinates of the points where the graph intersects the x-axis.



a. 1 and 2

b. -1 and -2

c. 2 and 3

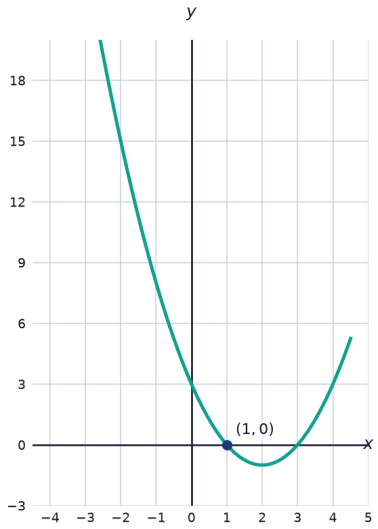
d. -1 and 2

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B. Determine the distance between A and B.



a. 2

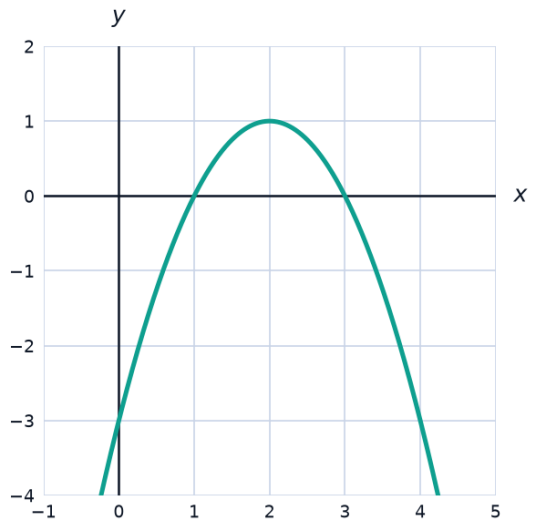
b. 3

c. 4

d. 5

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. Determine the value of $f'(2)$.



a. -2

b. 0

c. 2

d. 4

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. The vertex of the parabola is at point C. Determine the coordinates of the point where the graph intersects the y-axis.



a. (0, 3)

b. (3, 0)

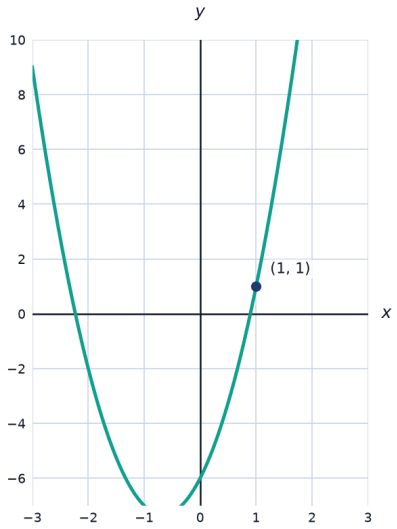
c. (0, -1)

d. (-1, 0)

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

The diagram shows the graph of the function $f(x) = 3x^2 + bx - 6$. The graph passes through the point $(1, 1)$. Find the value of b.



a. 3

b. 4

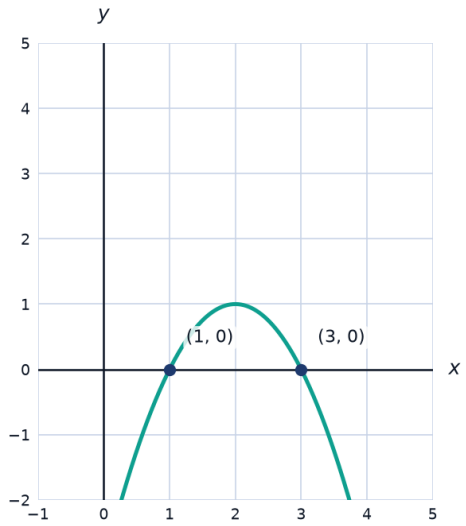
c. 5

d. 6

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Consider the function $f(x) = -x^2 + 4x - 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at two points. What is the value of the x-coordinate of the point of intersection with the larger x-value?



a. 1

b. 2

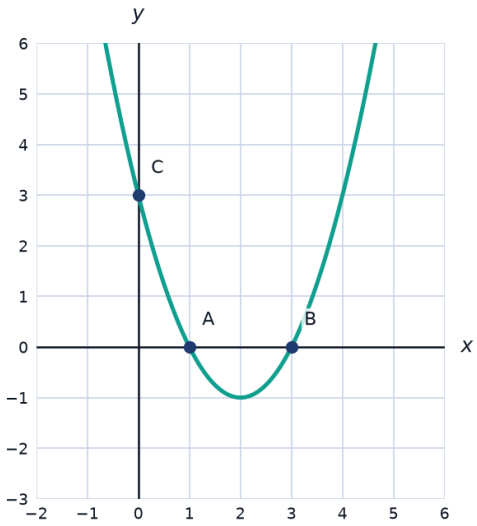
c. 3

d. 4

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.02 The Poisson Distribution

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B, and the y-axis at point C. Find the coordinates of the point where the graph of $y = f(x)$ intersects the y-axis.



a. (0, 3)

b. (3, 0)

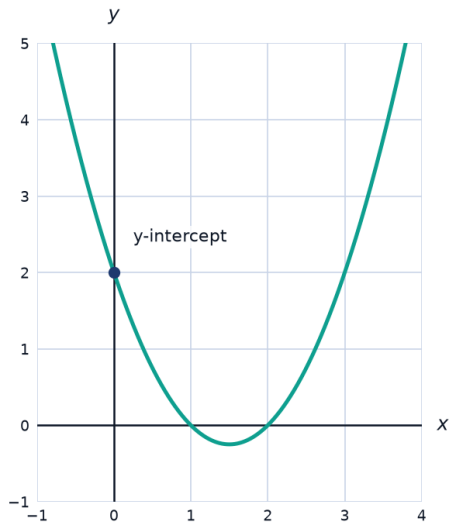
c. (1, 0)

d. (0, -3)

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Further Integration Techniques

Consider the function $f(x) = x^2 - 3x + 2$. The graph of $y = f(x)$ is shown. Determine the y-coordinate of the y-intercept.



a. 1

b. 2

c. 0

d. -1

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.02 Differential Equations

A particle's position in the complex plane is given by $z(t) = (2t^2 - 3t + 1) + i(t^3 - 6t^2 + 11t - 6)$, where t is time in seconds and $t \geq 0$. The particle starts at $t = 0$ and its path is traced by the complex number $z(t)$. The particle is said to be 'at rest' when its velocity, represented by the complex number $v(t) = z'(t)$, is zero. Determine the times $t > 0$ when the particle is at rest.

a. $t = 1$ and $t = 3$

b. $t = 1$ and $t = 2$

c. $t = 2$ and $t = 3$

d. $t = 1, t = 2,$ and $t = 3$

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